

InfiniteShader: Color Changeable 3D Printed Objects using Bi-Stable Thermochromic Materials

Yuto Umetsu
University of Tsukuba
Japan
umetsu@pml.slis.tsukuba.ac.jp

Parinya Punpongsanon
Osaka University
Japan
parinya@sys.es.osaka-u.ac.jp

Takefumi Hiraki
University of Tsukuba
Japan
hiraki@slis.tsukuba.ac.jp

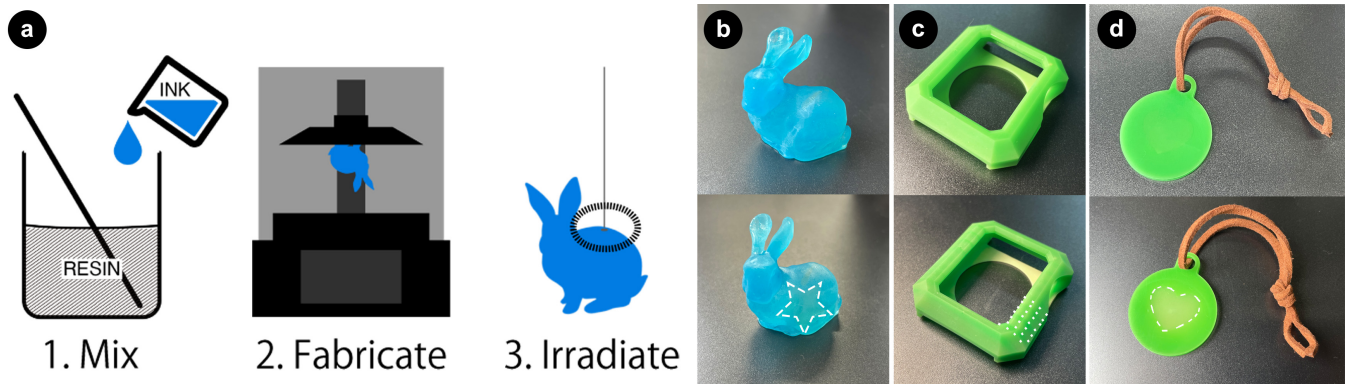


Figure 1: (a) the proposed fabrication process in which the bi-stable thermochromic materials are mixed with UV curable resin and directly printed from 3D printer. After printing the 3D object, we use the laser irradiation to modify the color and patterns of 3D printed objects. (b) - (d) the application scenario of InfiniteShader. We can control and change the patterns of (b) a figure, (c) a smartwatch cover, and (d) a bracelet after its fabrication. (white dashed lines are added for visibility.)

KEYWORDS

3d-printing, digital-fabrication, thermochromic materials

ACM Reference Format:

Yuto Umetsu, Parinya Punpongsanon, and Takefumi Hiraki. 2022. InfiniteShader: Color Changeable 3D Printed Objects using Bi-Stable Thermochromic Materials. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564209>

1 INTRODUCTION

Digital fabrication has been used to rapidly prototype designs and products. While the technology has the advantage of rapidly shaping ideas, the fabricated results usually cannot be changed its colors and properties after being printed. It is necessary to refabricate a new object to raise a new color and properties.

Previously addressed these issues utilizing special materials and fabrication method that enables control of the color and pattern of the structure after printing. ColorMod [Punpongsanon et al. 2018] and Photo-Chromeleon [Jin et al. 2019] utilized photochromic inks that emit and discharge the colors with light at specific wavelengths. Shader Printer [Saakes et al. 2012] proposed a method to control

patterns by applying thermochromic inks, which colors appear depending on the temperature, to an object and selectively heating it with a laser. However, these methods had problems in terms of color fading under natural light, low degree of freedom of structure, resolution, and time required for control.

In this study, we propose a method to control the color and pattern on the surface of 3D printed objects using bi-stable thermochromic materials. We mixed UV-curing resin with bi-stable thermochromic materials and selectively heated the surface of the 3D object with a laser (Fig. 1a). Unlike previous attempts that use similar bi-stable mechanism [Saakes et al. 2012], our method allows both 3D printed objects and later selectively change the color of 3D printed objects. In addition, we have fabricated various items using our proposed method and explored its application scenarios.

2 METHOD

The proposed method consists of a 3D printing process using resin containing bi-stable thermochromic inks and a partial deactivation process by heating using selective laser light irradiation.

In the 3D printing process, we mix UV-curing resin with bi-stable thermochromic inks to create a modeling object. We obtained the bi-stable thermochromic inks from erasable fluorescent pens (Frixion Light, PILOT). The inks have a coloring temperature of around -10°C and a decoloring temperature of 60°C or higher, the color can remain in the temperature range between -10°C to 60°C . Therefore, the inks rarely cause false coloring and decoloring due to unintended factors in everyday environments. In the process, the UV-curing resin (Clear UV resin, ANYCUBIC) and the bi-stable

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564209>

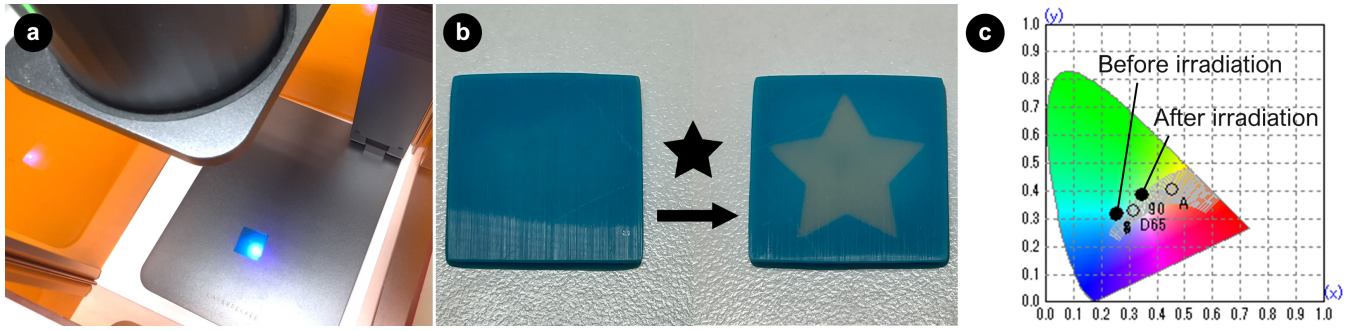


Figure 2: (a) the deactivation process for the 3D printed objects using a home laser processing machine. (b) the appearance change of the object before and after the deactivation process. A star pattern was drawn on the board-like object. (c) the chromaticity diagram of the object before and after laser irradiation.

thermochromic ink are mixed at a ratio of 10:1. Then, we fabricated a structure with a photolithographic 3D printer (Photon mono, ANYCUBIC) using the mixed resin. After printing, additional curing was performed using an external curing device (Form Cure, formLabs) with 395nm UV light irradiation at 70°C for 15 minutes. Then, the object is cooled down using a cooling spray (Safecool Check, Hozan) for the activation.

In the deactivation process, we selectively irradiate and heat the objects with a laser to deactivate the bi-stable thermochromic ink and draw our intended patterns. To scan the laser to draw the patterns, we import images of the patterns into an image editing software and generate a control code (G-Code). Then, we transfer the images to a personal laser processing machine (LaserPecker Pro, LaserPecker) using a smartphone application (LP2, LaserPecker), and the machine irradiates the laser onto the objects to heat and deactivate the thermochromic dye at the intended area (Fig. 2a).

3 EXPERIMENT

We conducted an experiment to measure chromaticity to evaluate the deactivation of the thermochromic dye on the 3D printed object. In this experiment, we compared the chromaticity of the objects with that after laser irradiation. We used a plate object (20mm × 20mm × 5mm) that was formed with the resin mixed with the blue bi-stable thermochromic ink, and irradiated the laser at a movement speed of 12 mm/s so that the trajectory drew a square 10 mm square edge. We selected three sample points from the surface of the object (the center point and two points on the laser trajectory) and measured the chromaticity values in the CIE XYZ color space using a spectroradiometer (SR-3AR, Topcon).

Figure 2b shows the appearance of the object after the laser irradiation. We can see that the area heated by laser irradiation became transparent; therefore, we can deactivate the thermochromic dyes. Figure 2c shows the xy chromaticity diagram before and after the laser irradiation. From these results, we confirmed that we could activate and deactivate the bi-stable thermochromic dyes on the surface of an object using selective heating by laser irradiation.

4 APPLICATION

We have explored the application scenarios of the proposed method by fabricating three different objects and controlling the color patterns. Figure 1b, 1c, and 1d show applications for a figure, a smart-watch cover, and a bracelet, respectively. We can modify the surface

patterns of these objects, so that we can enjoy the objects in different variations according to the situation and preference. In these applications, designers can also distribute the surface patterns and their processing codes as downloadable data so that users can easily update the appearance of the objects using their home laser processing machines. In addition, thanks to the bi-stability properties of thermochromic dyes, the objects' patterns will remain sharp and visible even in outdoor and other daylight environments.

5 CONCLUSION

In this paper, we proposed a method to control the color and pattern on 3D printed objects using bi-stable thermochromic inks. We mixed the UV-curing resin with bi-stable thermochromic ink and 3D printed the objects. It allows selective manipulation of color and pattern after printing. We experimented to evaluate the control of color loss in terms of chromaticity, and it was confirmed that the color and pattern of the printed material could be controlled by this method. Improving the coloring method in term of the accuracy is required as for the future work.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number JP21H05798, JP22H01447, JST ERATO Grant Number JPMJER1501, and HAYAO NAKAYAMA Foundation for Science, Technology and Culture.

REFERENCES

- Yuhua Jin, Isabel Qamar, Michael Wessely, Aradhana Adhikari, Katarina Bulovic, Parinya Punpongsanon, and Stefanie Mueller. 2019. Photo-Chromeleon: Re-Programmable Multi-Color Textures Using Photochromic Dyes. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)*. 701–712.
- Parinya Punpongsanon, Xin Wen, David S. Kim, and Stefanie Mueller. 2018. ColorMod: Recoloring 3D Printed Objects Using Photochromic Inks. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. 1–12.
- Daniel Saakes, Masahiko Inami, Takeo Igarashi, Naoya Koizumi, and Ramesh Raskar. 2012. Shader Printer. In *ACM SIGGRAPH 2012 Emerging Technologies (SIGGRAPH '12)*. Article 18, 1 pages.